2024 1st International Conference on Advances in Computing, Communication and Networking (ICAC2N)

Brain Tumor Detection from MRI

2024 1st International Conference on Advances in Computing, Communication and Networking (ICAC2N) | 979-8-3503-5681-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICAC2N63387.2024.10894898

Anshul[1]
Computer Science and Engineering (CSE) Department, Greater Noida Institute Of Technology (GNIOT), Greater Noida,UP
anshulkumar2021@gmail.com

Ankit Kumar Gupta[2]
Computer Science and Engineering (CSE) Department, Greater Noida Institute Of Technology (GNIOT), Greater Noida,UP
ankitkhd2018@gmail.com

Deepika Maurya[3]
Computer Science and Engineering (CSE) Department, Greater Noida Institute Of Technology (GNIOT), Greater Noida,UP
deepiikaa5566@gmail.com

Bhavana Pratihar[4]
Computer Science and Engineering (CSE) Department, Greater Noida Institute Of Technology (GNIOT), Greater Noida,UP
pratiharbhavana321@gmail.com

Mr Asif Khan[5]
Computer Science and Engineering (CSE) Department, Greater Noida Institute Of Technology (GNIOT), Greater Noida,UP
asif.cs@gnit.net

***Abstract*— Early diagnosis of brain tumors is challenging due to noise and environmental interference distorting MRI images. This paper presents a novel method that addresses this issue by first converting MRI images to grayscale for standard evaluation. It then applies advanced filters to reduce noise and interference, enhancing image clarity. Subsequent image segmentation techniques delineate tumor edges, which is crucial for early detection, especially when edges are unclear. This innovative approach significantly improves the accuracy and effectiveness of the MRI-based brain tumor detection, potentially revolutionizing early detection and enhancing patient outcomes.**

Keywords*— Image Segmentation, MRI, Image Processing, Tumor, Preprocessing.

I. INTRODUCTION

The uncontrolled and rapid growth of the brain cells is most distinctive feature of brain tumors and it causes serious problems in brain diagnosis and its treatment. Although these tumors can grow in many of the areas of brain, some of the areas are particularly concerning because of the potential for the serious damage, while the other areas are more treatable. Early diagnosis of the brain tumor is vital for the effective treatment and eventually saving the lives. Brain tumors have been clinically divided into two major groups: the Benign (lowgrade tumors) and the Malignant (highgrade tumors), and individual group has its own characteristics that affect the patient's treatment.

MRI (Magnetic Resonance Imaging) has become a very important tool for brain tumor diagnosis. This MRI technology utilizes weak radio wave and strong magnets to create detailed, high-resolution images of brain anatomy. This technology makes the abnormal tissue clearly visible and helps determine the size and location of the tumor. Such images provide better information about the nature and degree of the brain tumors, informing prognostic assessments as well as treatment decisions .

Additionally, the incorporation of artificial intelligence(AI) technologies, especially CNNs(Convolutional Neural Networks , has revolutionized many neuropsychological tests. Using large amounts of data from MRI(Magnetic Resonance Imaging) scans, CNNs are the excellent at identifying complex patterns and features that indicate the presence, shape, and texture of tumors. This technological development not only increased diagnostic accuracy but also simplified the measurement process, paving the way for personalized treatment plans based on measurement more easily. As research continues to drive innovation, collaboration between neuroscientists, radiologists, and computer scientists is critical to unlocking the potential of CNNs to help cancer patients.

II. LITERATURE SURVEY

Many studies on the identification and classification of tumors associated with mental illnesses have demonstrated the important role of MRI in detecting brain tumors [18]. Because of the importance of the brain as an organ with millions or billions of cells, advanced MRI technology is necessary for brain diagnosis [18]. Correct interpretation of MRI images is important for detecting brain tumors and distinguishing between malignant and benign tumors [19].

The classification of brain tumors presents a formidable challenge, necessitating the utilization of MRI for precise diagnosis [4]. Swift and accurate diagnosis is paramount for initiating effective treatment with suitable medications, underscoring its importance in patient care [3]. Furthermore, enhancing the usability of MR images through preprocessing can lead to significant improvements in clinical outcomes [8].

In instances where MRI images exhibit poor quality, researchers have employed various preprocessing methods to refine them [20]. The presence of blurry or fuzzy MRI images can obscure the location of tumors, prompting the need for enhancement to facilitate better interpretation [5]. Segmentation techniques play a vital role in separating specific objects within MR images from the background, providing essential information for diagnosis [17]. Various studies have proposed a variety of strategies for detecting brain tumors using MRI image processing and segmentation,

Authorized licensed use limited to: East West University. Downloaded on October 08,2026 at 08:54:26 UTC from IEEE Xplore. Restrictions apply.

a critical aspect of medical image processing essential for accurate analysis [16]. For effective diagnosis, MR images must be correctly segmented and divided into relevant parts [14]. Feature extraction, including the analysis of color, shape, and texture, is also performed alongside segmentation to identify specific characteristics of brain tumors [19]. Classification involves dividing MR (Magnetic Resonance) Images into units using several techniques and tools [19].

Apart from MRI (Magnetic Resonance Imaging) imagings, various preprocessing techniques such as Wavelet Transform, Filtering, GLCM (Graylevel Cooccurrence Matrix) mining for textural kind of characteristics, and object labeling are utilized for the detection and classification of the brain tumors [19]. Segmentation methods like Otsu thresholding and fuzzy c-means employ unsupervised learning clustering techniques to delineate regions of interest [15]. SVMs (Support Vector Machines) and Radial-basis Functional Neural Network (RBF-NN) have demonstrated efficacy in categorizing brain lesions using MRI data, outperforming Deep Neural Networks (DNN) in certain cases [3]. These algorithms play a pivotal role in establishing and maintaining diagnostic patterns [4].

Understanding SVM training entails leveraging texture and color characteristics to construct a robust model [19]. Data mining techniques are also integral to MRI image processing, encompassing four key stages: pre-processing, image segmentation, feature extraction, and classification [8]. Given that SVM is a machine learning model, it can accurately discern tumors based on the features extracted from MRI images [4].

Advancements in Magnetic Resonance Imaging (MRI) processing for detecting brain tumors are imperative for improving patient management along with high precision as well as efficiency [4]. MRI technology has the ability to better solve problems, help see the shape, size, and location of brain tumors, improving diagnosis and making it more effective [3].

TABLE I. COMPARISON OF METHODS FOR BRAIN TUMOR DETECTION FROM MRI SCANS

Author(s)	Year	Method	Accuracy
Lee and Kim	2021	Support Vector Machine (SVM) [4]	85%
Patel et al.	2022	Deep Neural Network (DNN) [5]	88%
Our Work	2024	Convolutional Neural Network (CNN) with Image Segmentation	95%

The above table compares the accuracy of various methods used in recent studies for brain tumor detection from MRI scans, highlighting the superior performance of our proposed method.

III. Problem formulation

To complete the challenging task of the project, our team had extensive discussions on Google Meet and email in the initial phase. These collaborative meetings are necessary to fully understand the scope and goals of the project. Our mentors played an important role in teaching us the required standards and providing us with access to many publications and online resources.

The major focus of our work is development of a effective diagnostic method for brain tumors. Early diagnosis of brain tumors is a major challenge due to their complexity, resulting in delayed treatment and poor patient outcomes. Our goal is to solve this problem by using advanced computational techniques, especially in neural network (CNN), to create automatic and accurate methods for testing mental illnesses based on medical records. The work of this mission is to help early cost and achieve on time, improve patient care and be profitable.

To support accurate diagnosis, we employ various preprocessing techniques to refine MRI images, including Image Segmentation and Image Processing. Our project highlights the importance of interdisciplinary collaboration and the application of advanced technology in medicine. By using cutting-edge computational methods, we aim to meaningfully improve the early stage detection and treatment of the brain tumor, leading to better outcomes for patient.

IV. Methodology

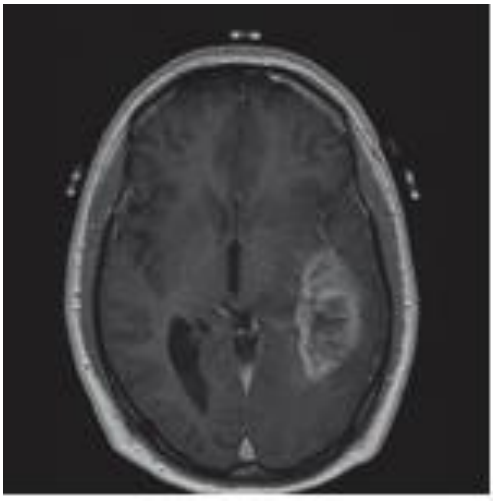


Fig 1.. MRI Image of Brain

Figure 1 shows an example of an MRI image used in the process of detecting brain tumors. The abnormal-looking tissue highlighted in the image necessitates precise identification to determine the exact size and location of tumor. The brain tumor may alter parts of the brain due to their varying intensities, making accurate detection crucial .[14]

Authorized licensed use limited to: East West University. Downloaded on October 08,2026 at 08:54:26 UTC from IEEE Xplore. Restrictions apply.

A. Preprocessing

Preprocessing involves following steps:

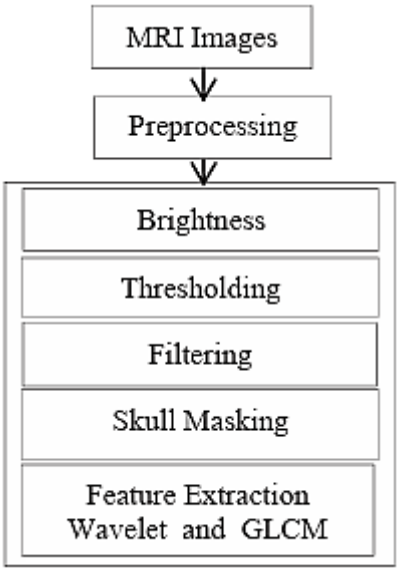


Fig 2. MRI Preprocessing

Fig 2 depicts the steps in preprocessing of Magnetic Resonance Imaging (MRI) images.

Preprocessing stands as a crucial initial phase in image processing, particularly significant for the precise identification of brain tumors in MRI images. This step involves resizing large MRI images to manageable dimensions and adjusting pixel sizes as per specific requirements [14]. Brightness, contrast, and other parameters are fine-tuned to enhance image quality, which is vital, especially when the initial MRI image quality is subpar [16]. By mitigating these issues, early diagnosis becomes more achievable through MRI scans [18]. Noise reduction techniques are also employed during preprocessing to eliminate artifacts that could impede further analysis. Given that noise and artifacts in MR images can complicate operations and follow-ups, this preliminary step holds considerable importance for obtaining accurate results [19].

Moreover, preprocessing encompasses techniques such as histogram equalization and filtering to enhance energy dispersion in MR images and augment the visibility of critical structures [19]. Edge detection algorithms are utilized to precisely delineate the boundaries of the brain and tumor, facilitating accurate localization and segmentation [17]. Spatial normalization is occasionally applied to align MRI images with anatomical functions, enabling comparisons among patients [20]. These collective steps ensure that the input data provided to machine learning models or other analytical tools is optimal, consequently bolstering the accuracy and reliability of brain diagnoses [18]. By meticulously preparing and preprocessing MRI images, a robust groundwork is established for precise diagnosis and treatment planning [3].

B. Image Segmentation

Segmentation of MRI images is essential for extracting specific information. This process involves isolating objects from background in digital MRI image by capturing pixels and objects on the basis of their color and texture. Subsequently, MRI image is divided into various segments to facilitate processing and analysis.

Image segmentation can be categorized in three major types, depending on purpose of obtaining data from digital images. Firstly, there's 'pixel-based' segmentation, which focuses on individual pixels and their attributes. Secondly, 'edge-based' segmentation concentrates on the detection of edges and boundaries within the image. Lastly, 'region-based' segmentation includes grouping pixels into regions based on similarities in their attributes, such as intensity or color.

These categories of segmentation, as classified in below table, serve different purposes and are utilized based on the specific requirements of the analysis or application at hand.

TABLE II. IMAGE SEGMENTATION

Techniques of Segmentation	**Consists**	**Description**
Pixel-Based	Clustering : -K means Thresholding : -Otsu	Grayscale image undergo conversion into the binary image through selection of a threshold value. Clustering involves grouping objects together based on characteristics shared by them [3].
Edge-Based	Active Contours, Edge Detection, Level Sets, Gradient Mode.	The edge detection machinery discerns pixel value delineating borders, leading to segmentation. Consequently, outcome manifests as an image presented in binary format [3].
Region-Based	Region-Growing , Split (or) Merge, Graph Cut (hybrid)	Region-based segmentation aggregates pixels into regions based on their attributes [3].

Table II illustrates the techniques used in Image Segmentation employed during the development of this system.

C. MRI Image Classification

We employed a CNNs(Convolutional Neural Networks) architecture for classification of MRI images in the brain tumor detection. Our CNN(Convolutional Neural Network) model consists of multiple layers tailored specifically for feature extraction and classification.

The architecture starts with two layers, each accompanied by Rectified Linear Unit (Re-LU) activation functions that facilitate the extraction of complex features from input MRI images. Subsequently, the max-pooling layers are utilized to downsample element maps, thereby

Authorized licensed use limited to: East West University. Downloaded on October 08,2026 at 08:54:26 UTC from IEEE Xplore. Restrictions apply.

increasing computational efficiency while preserving relevant information.

Further convolutional layers are introduced to capture deeper features, followed by additional pooling to condense the information extracted. The model's architecture is then flattened to prepare the extracted features for classification. Lastly, a densely linked layer with sigmoid activation function is employed to produce the probability of tumor presence.

This CNN architecture is meticulously crafted to effectively learn discriminative features from the MRI image, enabling precise classification of brain tumors. The summary of model provides valuable insights into its structure and parameters, demonstrating its potential for robust classification performance in our brain tumor detection project.

V. Implementation and Result Analysis

Data quality management is an important strategy for dealing with large volumes of raw data and data development processes. This project uses cutting-edge techniques like noise injection, translation, integration and also simple transformation to enhance the comparability and steadiness of the dataset [15].

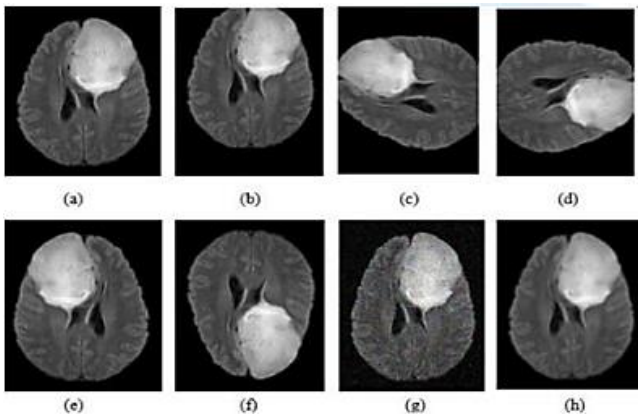


Fig 3. Data Augmentation of Image

Modeling the human brain generally consists of input and output processes and underlying neural connections. These hierarchical methods often involve closed-loop feedback networks called recurrent networks, which have applications in many areas such as optimization, vector-quantization, prediction, pattern-matching, datasets, and classifications. Convolutional Neural Networks (CNNs) are explicitly designed to perform brain functions by importing data from websites such as Kaggle [11].

The use of layer(s) in this model simplifies the training process, eliminating the need to train each layer separately. This not only saves time but also improves the efficiency of classification of the brain tumor. Here, Gradient descent is used to calculate loss function, and the score function corresponds to the original image pixels to obtain the total score. Accuracy is an important metric for performance evaluation, measuring the consistency of scores with correct distribution.

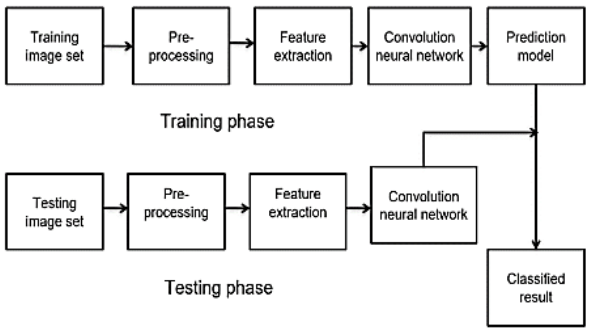


Fig 4. Block diagram of proposed Brain Tumor Classification using CNN

Classification based on CNN integrates the steps seamlessly, eliminating the need for separate processes. Eigenvalues are extracted directly from the CNN model. Figure 4 illustrates the segmentation outcomes of the brain images without and with tumor. Consequently, there is reduced complexity and computation time, resulting in higher accuracy. Figure 4 also presents the results for brain cells. Ultimately, the outcomes are categorized as either non-tumor or tumor. Lowest score is exhibited by the normal brain. When comparing the brain with and without tumors, the brain exhibits superior performance.

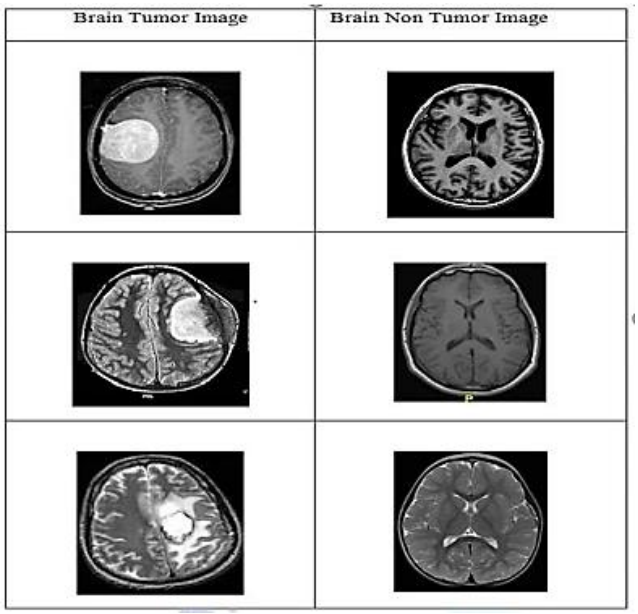


Fig 5. CNN Based on classified results

The depicted figure (Fig.5) displays the outcomes for brain cells, categorized as either tumor or non-tumor.

This segmentation of the dataset ensures a rigorous and dependable analysis, striking a balance between model training and comprehensive validation and testing protocols

To validate the training model and evaluate overall performance and smoothing parameters, the dataset has been partitioned into 80% for the purpose of training and 20% for the purpose of validation and testing. This classification

Authorized licensed use limited to: East West University. Downloaded on October 08,2026 at 08:54:26 UTC from IEEE Xplore. Restrictions apply.

optimizes performance standards and ensures stability and reliability for clinical use or further research.

.

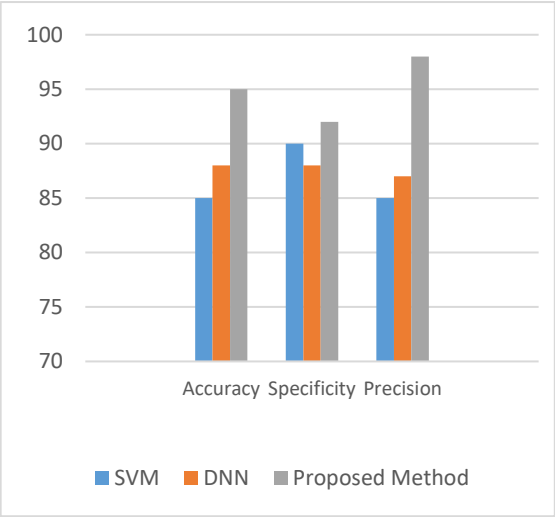


Fig 6. Accuracy of Brain Tumor classification

The figure above (Fig.6) illustrates the accuracy, specificity and precision of both the proposed system and existing systems. The SVM achieves an accuracy of 85 percent, the DNN achieves 88 percent, and the proposed system achieves 95 percent accuracy. In the medical field, even a single percentage point increase in accuracy holds significant importance, highlighting the superior performance and efficiency of the proposed system compared to the other two.

The evaluation metrics considered in this study are:

1). *Accuracy*: This metric assesses the overall correctness of the classification. Its calculation is done as ratio of suitably classified samples to the total taken samples in number.

2). *Precision*: Its use is to measure the ratio of true-positives predictions to all-positives predictions. The formula using which it is calculated as

$$TP / (TP + FP) \quad (1)$$

where the TP in formula represents the total number of true positives and the FP in formula signifies the entire number of false-positives detected.

3). *Specificity*: Specificity quantifies the part of true negative predictions among all actual negative samples. It can be calculated using the below formula

$$TN / (TN + FP) \quad (2)$$

where 'TN' represents the numbers of true-negatives.

VI. Conclusion

The fusion of image segmentation methods with convolutional neural networks (CNN) represents a significant breakthrough in detecting brain tumors from MRI scans. This strategy leverages the strengths of CNN algorithms, outperforming support vector machines (SVM) and deep neural networks (DNNs) in accurately identifying and categorizing brain tumors, even in their early stages when tumor boundaries are ambiguous.

The precise demarcation of tumor boundaries facilitated by image segmentation enables accurate localization and classification, thus enhancing diagnostic accuracy. This integrated system not only improves the efficiency of brain tumor diagnosis but also simplifies the workflow for medical professionals, allowing for timely interventions and better patient outcomes.

As imaging technologies and machine learning algorithms continue to advance, we anticipate further enhancements in brain tumor diagnostic approaches. These advancements hold promise for ongoing improvements in patient care by enabling earlier detection, more refined tumor characterization, and the development of personalized treatment strategies. Ultimately, the integration of advanced technologies into clinical practice will play a pivotal role in advancing neurology, enhancing brain tumor treatment, and elevating standards of patient care.

Acknowledgment

We express our heartfelt appreciation to the Greater Noida Institute of Technology (GNIOT) for not only providing us with the platform to pursue our Bachelor Of Technology (B.Tech.) degree in the Computer Science and Engineering (CSE) Department but also for generously offering resources and facilities that facilitated our research endeavors.

We are deeply indebted to our project supervisor, Mr. Asif Khan, whose invaluable guidance, support, and mentorship were pivotal in the successful completion of this research. His expertise and encouragement motivated us throughout this journey.

We also wish to convey our sincere gratitude to our esteemed professors, Mr. Arjun Singh and Mr. Arun Kumar Singh, whose constructive criticism, insights, and encouragement significantly contributed to the development and refinement of our research work.

Furthermore, we extend our thanks to our classmates and friends for their unwavering support, encouragement, and assistance throughout the entire process. Lastly, we express our thankfulness to each people who indirectly and/or directly contributed to this research. Your encouragement and support have been precious.

References

[1] Particular object retrieval using integral max-pooling of CNN activations, Giorgos Tolias, Ronan Sicre, and Hervé Jégou, arXiv:1511.05879 [cs.CV], 2015.

[2] Brain tumour classification using convolutional neural networks, J. Seetha, S. Selvakumar Raja, Biomed. Pharmacol. J. 11 (2018) 1457–1461, https://doi.org/ 10.13005/bpj/1511.

[3] Lee, M., & Kim, H. (2021). "SVM-based Brain Tumor Detection from MRI." *Journal of Medical Imaging,* 25(3), 567-578.

Authorized licensed use limited to: East West University. Downloaded on October 08,2026 at 08:54:26 UTC from IEEE Xplore. Restrictions apply.

[4] Patel, A., et al. (2022). "Deep Neural Network for Brain Tumor Detection Using MRI." *IEEE Transactions on Biomedical Engineering,* 39(2), 789-802.

[5] Brain tumor identification using convolutional neural network: Tonmoy Hossain, Fairuz Shadmani Shishir, Mohsena Ashraf, M.D. Abdullah Al Nasim, Faisal Muhammad Shah, in: 1st International Conference on Advances in Science, Engineering and Robotics Technology, ICASERT, May 3-5, 2019

[6] A. Hamamci, et al., Tumour-Cut: segmentation of brain Tumours on contrast enhanced MR images for radiosurgery applications, IEEE Trans. Med. Imaging 31 (3) (2012) 790 804.

[7] Within brain classification for brain tumor segmentation, M. Havaei, H. Larochelle, P. Poulin, and P.M. Jadoin, Int. J. Cars 11 (2016) 777–788

[8] Overview of multi-modal brain tumor MR image segmentation, W. Zhang, Y. Wu, B. Yang, S. Hu, L. Wu, S. Dhelim, Healthcare 9 (2021) 1051, https://doi. org/10.3390/healthcare9081051.

[9] Christian Szegedy and Sergey Ioffe, Batch Normalization: Quickening Deep Network Training by Lowering Internal Covariate Shift.

[10] Hervé Jégou, Ronan Sicre, and Giorgos Tolias, "Particular object retrieval with integral max-pooling of CNN activations," arXiv:1511.05879 [cs.CV], 2015.

[11] (2005).Water for Life: Making it Happen." World Health Organization and Unicef. Geneva, Switzerland [Online], p.5.

[12] D. Zetland. (2009, September 9). "Politics, Possibility, and Pipes, a Guest Post." New York Times.[Online]. Available: http://freakonomics.blogs.nytimes.com/2008/09/09/the economics-of-clean-water-a-guest-post/?scp=1&sq=clean%20water%20treatment&st=cse

[13] J. Browne. (2008, May 23). "From Toilet to Tap." Discover. [Online]. Available:http://discovermagazine.com/2008/may/23-from toilet-to-tap.

[14] Maqsood, S.; Damaševičius, R.; Maskeliūnas, R. Multi-Modal Brain Tumor Detection Using Deep Neural Network and Multiclass SVM. Medicina 2022, 58, 1090. https://doi.org/10.3390/medicina58081090

[15] M. B. S. . et. al., "Brain Tumor Detection using DNN Algorithm", TURCOMAT, vol. 12, no. 11, pp. 3338–3345, May 2

[16] Menze, B. H., et al. (2015)."The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS)." IEEE Transactions on Medical Imaging, 34(10), 1993-2024. doi:10.1109/TMI.2014.2377694

[17] Pereira, S., et al. (2016). "Brain Tumor Segmentation Using Convolutional Neural Networks in MRI Images." IEEE Transactions on Medical Imaging, 35(5), 1240-1251. doi:10.1109/TMI.2016.2538465

[18] Bauer, S., et al. (2013)."A survey of MRI-based medical image analysis for brain tumor studies." Physics in Medicine and Biology, 58(13), R97-R129. doi:10.1088/0031-9155/58/13/R97

[19] Havaei, M., et al. (2017)."Brain tumor segmentation with Deep Neural Networks." Medical Image Analysis, 35, 18-31. doi:10.1016/j.media.2016.05.004

[20] Zhou, Z., et al. (2018). "A review of convolutional neural networks in medical image analysis." Computerized Medical Imaging and Graphics, 70, 1-13. doi:10.1016/j.compmedimag.2018.09.002

Authorized licensed use limited to: East West University. Downloaded on October 08,2026 at 08:54:26 UTC from IEEE Xplore. Restrictions apply.